UGANDA INDUSTRIAL RESEARCH INSTITUTE

Namanve Industrial and Business Park, Mukono District, Uganda

SYSTEM DESIGN DIAGRAMS

FOR THE DESIGN AND DEVELOPMENT OF A WEB BASED

INVENTORY MANAGEMENT SYSTEM

FOR UGANDA INDUSTRIAL RESEARCH INSTITUTE (UIRI)

NAKAWA AND NAMANVE CAMPUSES

Name	Registration Number	Project Responsibility
Keith Paul Kato	24/U/26593/EVE	Lead Reporter and Database Design
Komukama Tracy	24/U/06151/EXT	Requirements and Documentation
Atuhiire Deo Kamate	24/U/14365/PS	System Analysis and Testing
Programme	Bachelor of Science in Computer Science	
Institution	Makerere University, Kampala and Jinja	
Host Organisation	Uganda Industrial Research Institute (UIRI)	
Department	Information and Communications Technology (ICT)	
Field Supervisor	Mr. William Kakooza: IT Engineer at UIRI	
Submitted to	ICT Team, UIRI Namanve	
Submission Date	July-August 2026	

Document Control

This document presents the system design diagrams for the UIRI Inventory Management System. It converts the approved proposal, SRS and database design into visual models that can guide implementation, testing and stakeholder review.

Table 1 Document Control Details

Document Title	System Design Diagrams for the UIRI Inventory Management System
Project	Design and Development of a Web Based Inventory Management System for UIRI
Version	1.1
Prepared For	ICT Team, UIRI Namanve
Prepared By	Keith Paul Kato, Komukama Tracy and Atuhiire Deo Kamate
Document Status	Revised Landscape Version

Revision History

Table 2 Revision History

Version	Date	Description	Prepared By
1.0	July-August 2026	Initial system design diagrams prepared after database design approval	Project Team
1.1	July-August 2026	Reworked into landscape format with clearer figures, extended arrows, visible labels and linked contents	Project Team

Table of Contents

List of Figures

List of Tables

Abbreviations and Acronyms

Table 3 Acronyms and Abbreviations

Acronym	Full Form
DBMS	Database Management System
ERD	Entity Relationship Diagram
ICT	Information and Communications Technology
IMS	Inventory Management System
RBAC	Role Based Access Control
SRS	Software Requirements Specification
UML	Unified Modeling Language
UIRI	Uganda Industrial Research Institute

1.0 Introduction

This document presents the system design diagrams for the proposed UIRI Inventory Management System. It follows the System Proposal, Software Requirements Specification and Database Design Document. The purpose is to show how users will interact with the system, how major workflows will proceed, and how the main software classes will support the database driven implementation.

The document focuses on the diagrams required in the project brief: Use Case Diagram, Activity Diagrams and Class Diagram. The diagrams are prepared in a Visio style so that they can be redrawn or refined in Microsoft Visio without changing the underlying design logic.

2.0 System Design Scope

The system design covers authentication, role based authorization, campus and department setup, inventory item registration, fixed asset tracking, consumable stock management, supplier records, stock receipt, stock issue, stock transfer, maintenance recording, dashboard reporting and audit logging.

The initial pilot is expected to run at UIRI Namanve, while the design remains capable of supporting both Nakawa and Namanve campuses. The diagrams separate user interaction design from database design. The ERDs define how tables relate, while the UML diagrams in this document define how actors, workflows and software classes cooperate during system use.

The design also preserves a distinction between durable assets and consumable stock. Durable assets such as computers, machines and laboratory instruments are traced individually by serial number, condition and custodian. Consumables such as stationery, reagents and workshop materials are tracked by quantity and reorder levels.

3.0 System Context and Architecture

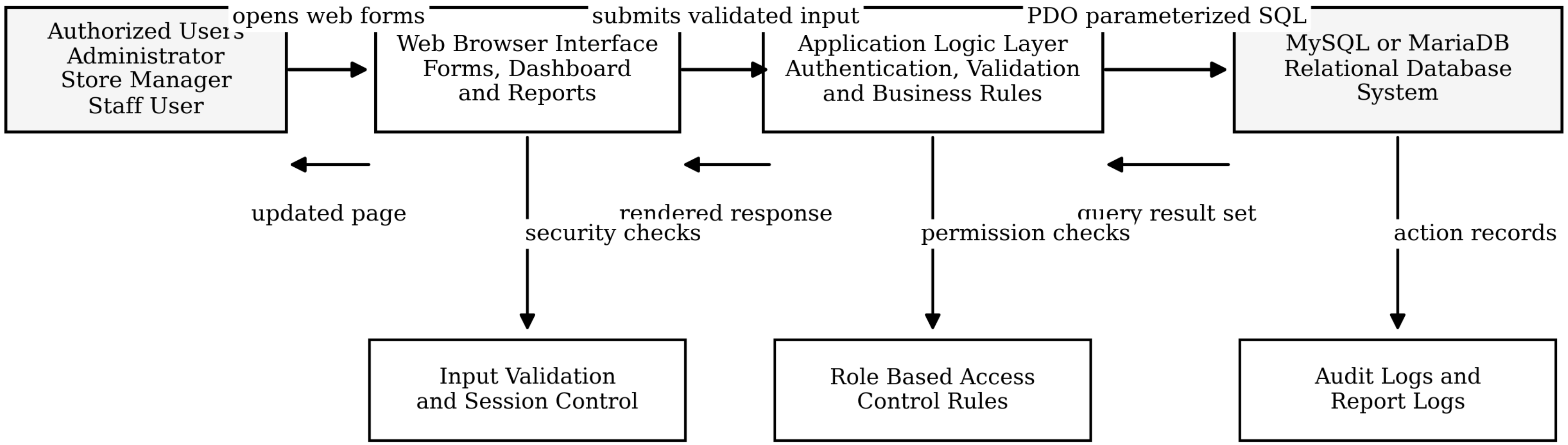
The proposed IMS follows a simple three layer arrangement. Users interact with forms, dashboards and reports through a web browser. The application logic layer validates input, checks permissions and applies business rules. The database layer stores users, departments, items, assets, stock transactions, requisitions, maintenance records and audit logs.

Figure 1 shows the request path, database interaction and response path. The arrows have been extended and labelled to make the data movement clear. A user action begins in the browser, passes through the application layer, reaches the database through parameterized SQL, and returns results to the user interface.

Table 4 System Architecture Responsibilities

Layer	Responsibility
User and Browser Layer	Provides forms, dashboards, report views and authenticated user interaction
Application Logic Layer	Validates input, checks role permissions, applies inventory business rules and writes audit records
Database Layer	Stores master data, item records, fixed assets, stock balances, transactions, requisitions and logs
Security Controls	Protect login sessions, validate data, restrict unauthorized actions and preserve accountability

System Architecture Context for the UIRI Inventory Management System



The arrows show the full request, processing, storage and response path used by the proposed web based inventory system.

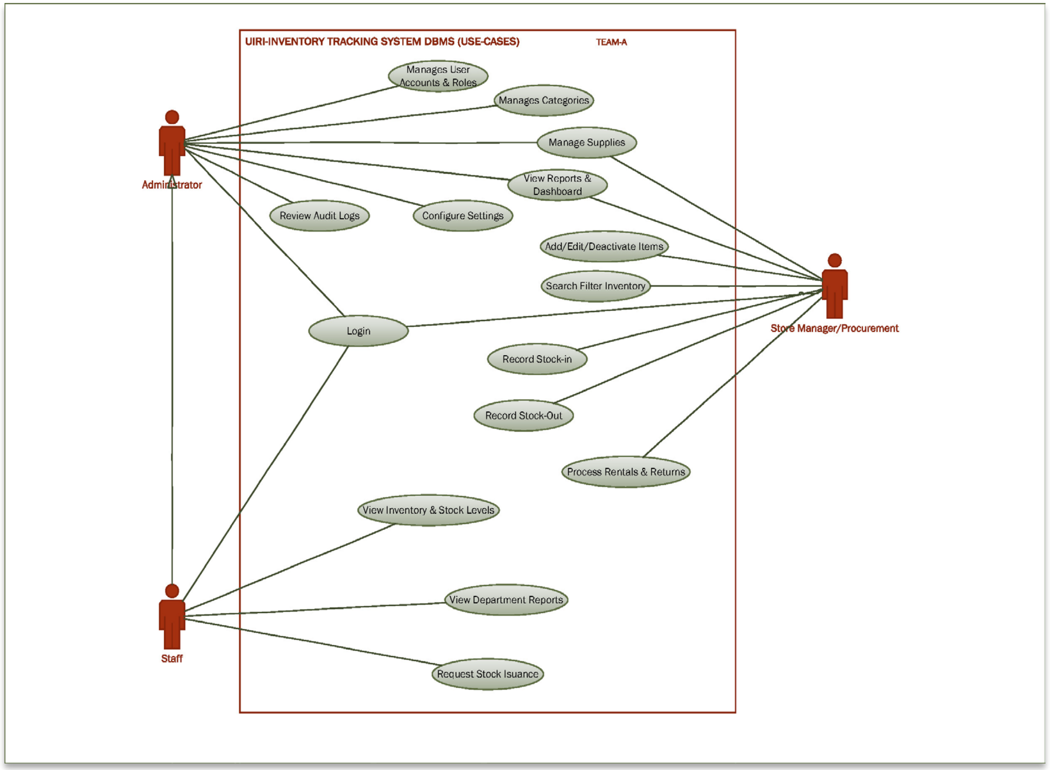
Figure 1 System Architecture Context

4.0 Actors and Use Case Model

The use case model identifies the external users of the system and the services each user expects from the IMS. These actors represent roles rather than individual people. A single physical person may perform more than one role only if the ICT Team grants the required permissions.

Table 5 Actor Summary

Actor	Description	Main Responsibilities
Administrator	Technical user with highest system privileges	Manage users, roles, campuses, departments, categories, system settings and audit review
Store Manager	Inventory custodian responsible for daily stock operations	Register items, receive stock, issue stock, transfer items, manage suppliers and maintain balances
Staff User	Departmental user who needs inventory information or items	View authorized records, submit requisitions and check request status
Department Head	Approving officer for departmental requests	Review, approve or reject requisitions and view departmental reports
Finance / Procurement	User concerned with suppliers and acquisition records	View supplier history, purchase references, batches and procurement related reports
ICT Supervisor	System owner or reviewer from the ICT function	Review dashboards, audit logs and technical accountability reports



Use Case Diagram for the UIRI Inventory Management System

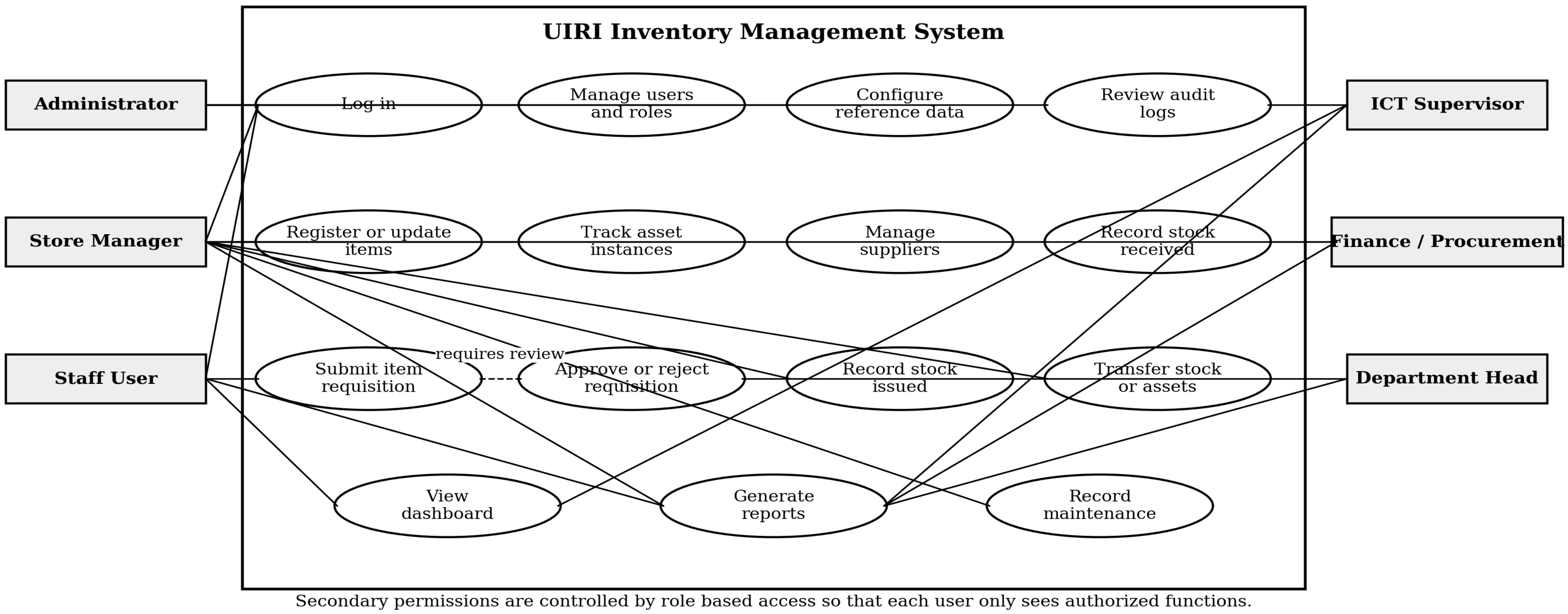


Figure 2 Use Case Diagram for the UIRI Inventory Management System

5.0 Use Case Specifications

The table below summarizes the main system use cases and provides a practical description for implementation and testing. More detailed use case narratives can be derived from these specifications during coding.

Table 6 Use Case Summary

Use Case	Primary Actor	Description	Expected Result
Log in	All users	User enters credentials and the system validates account status and role	Authenticated user is taken to the correct dashboard
Manage users and roles	Administrator	Administrator creates users, assigns roles and disables inactive accounts	Only authorized users can access the system
Register inventory item	Store Manager	Store Manager records item code, name, category, type, brand, model and reorder level	Item becomes available in the inventory catalogue
Track asset instance	Store Manager	Fixed assets are captured with serial number, location, status, purchase year and custodian	Each durable asset can be traced individually
Record stock received	Store Manager	Received quantity, supplier, batch and location are recorded	Stock balance increases and audit record is created
Record stock issued	Store Manager	Approved request is issued to a department or user	Stock balance decreases and issue history is saved
Generate reports	Authorized users	User selects report filters and report type	System displays or exports accurate filtered records
Review audit logs	Administrator or ICT Supervisor	System action history is reviewed for accountability	Sensitive actions can be traced to responsible users

Table 7 Core Use Case Specifications

Use Case	Precondition	Main Flow	Postcondition
Record Stock Received	User is logged in as Store Manager and item exists in catalogue	Select item, supplier, location and quantity. Enter batch and date. Save receipt.	Stock balance is increased and stock transaction is recorded
Record Stock Issued	Request is approved and stock is available	Select approved request or issue form. Confirm item and quantity. Save issue.	Stock balance is reduced, asset custodian may be updated and audit log is written
Generate Low Stock Report	User has report permission	Open reports, choose low stock report, apply campus or category filters.	System displays items whose available quantity is at or below reorder level
Update Asset Condition	Fixed asset exists in the system	Search asset, select condition and enter remarks.	Asset condition is updated and change is logged

6.0 Activity Diagrams

Activity diagrams show the flow of control for key system processes. They help the project team and stakeholders confirm that the proposed system supports practical inventory work before implementation begins.

The following diagrams are intentionally placed one per page so that each process flow remains visible when printed or reviewed on screen. Decision branches are labelled with Yes and No outcomes, while longer arrows show the direction of control between activities.

Table 8 Activity Diagram Summary

Activity Diagram	Purpose
Login and Role Based Access	Shows authentication, credential validation, active account checking and dashboard routing
Record Stock Received	Shows the process of receiving consumable stock or registering fixed asset instances
Requisition and Stock Issue	Shows request review, approval, stock availability checking and issue recording
Dashboard and Report Generation	Shows filtering, access checking, data retrieval and report output

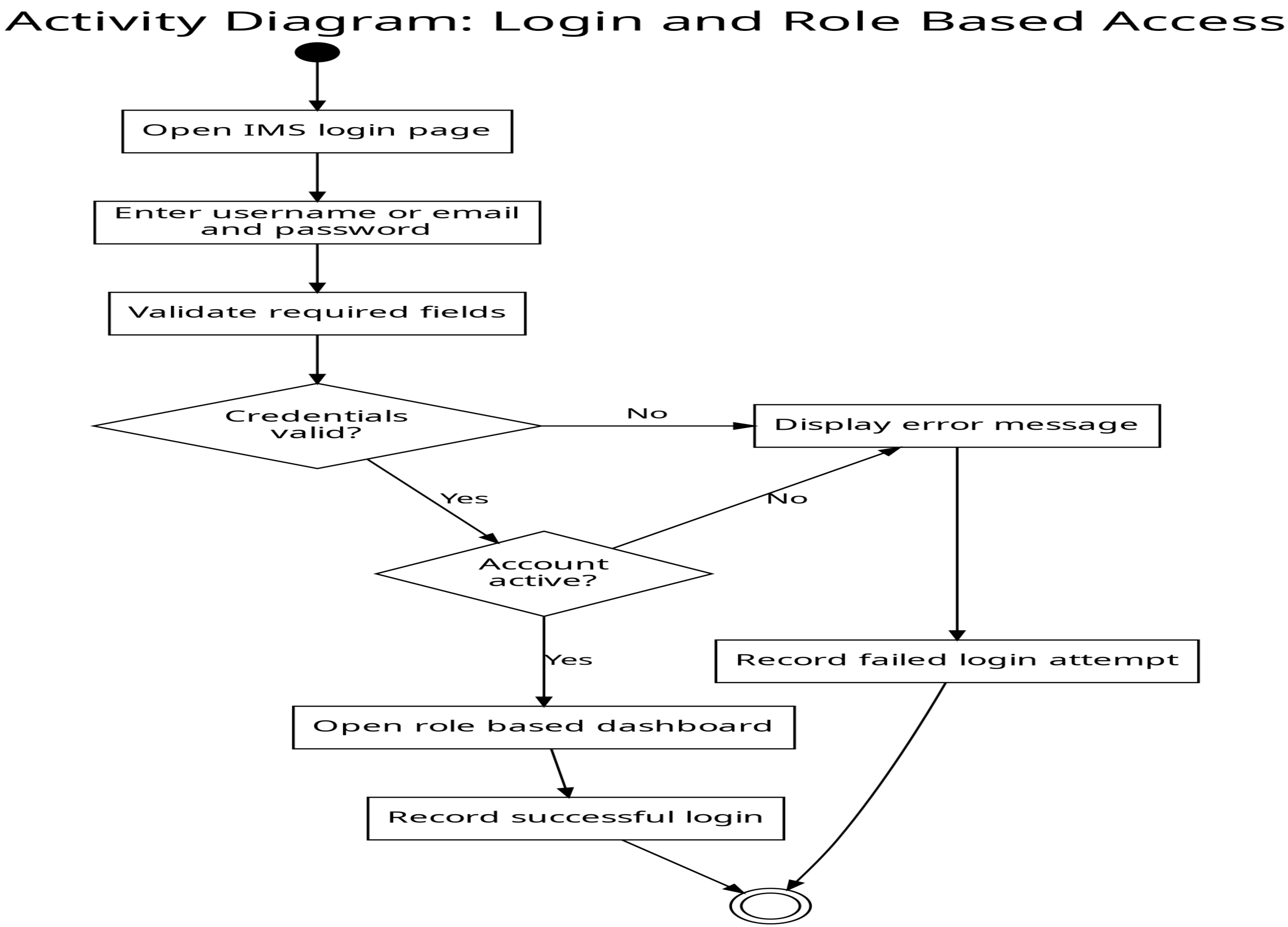


Figure 3 Activity Diagram for Login and Role Based Access

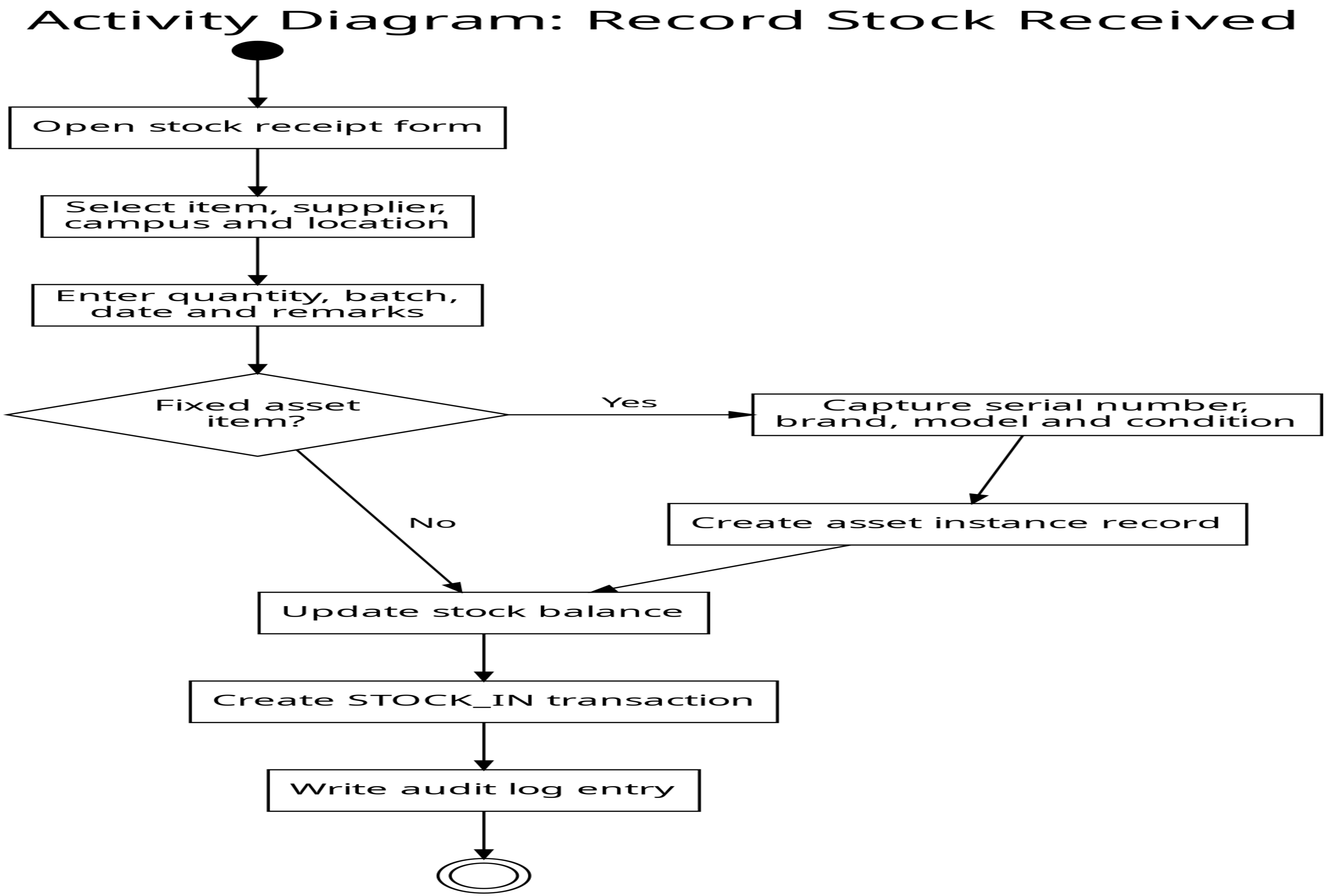


Figure 4 Activity Diagram for Recording Stock Received

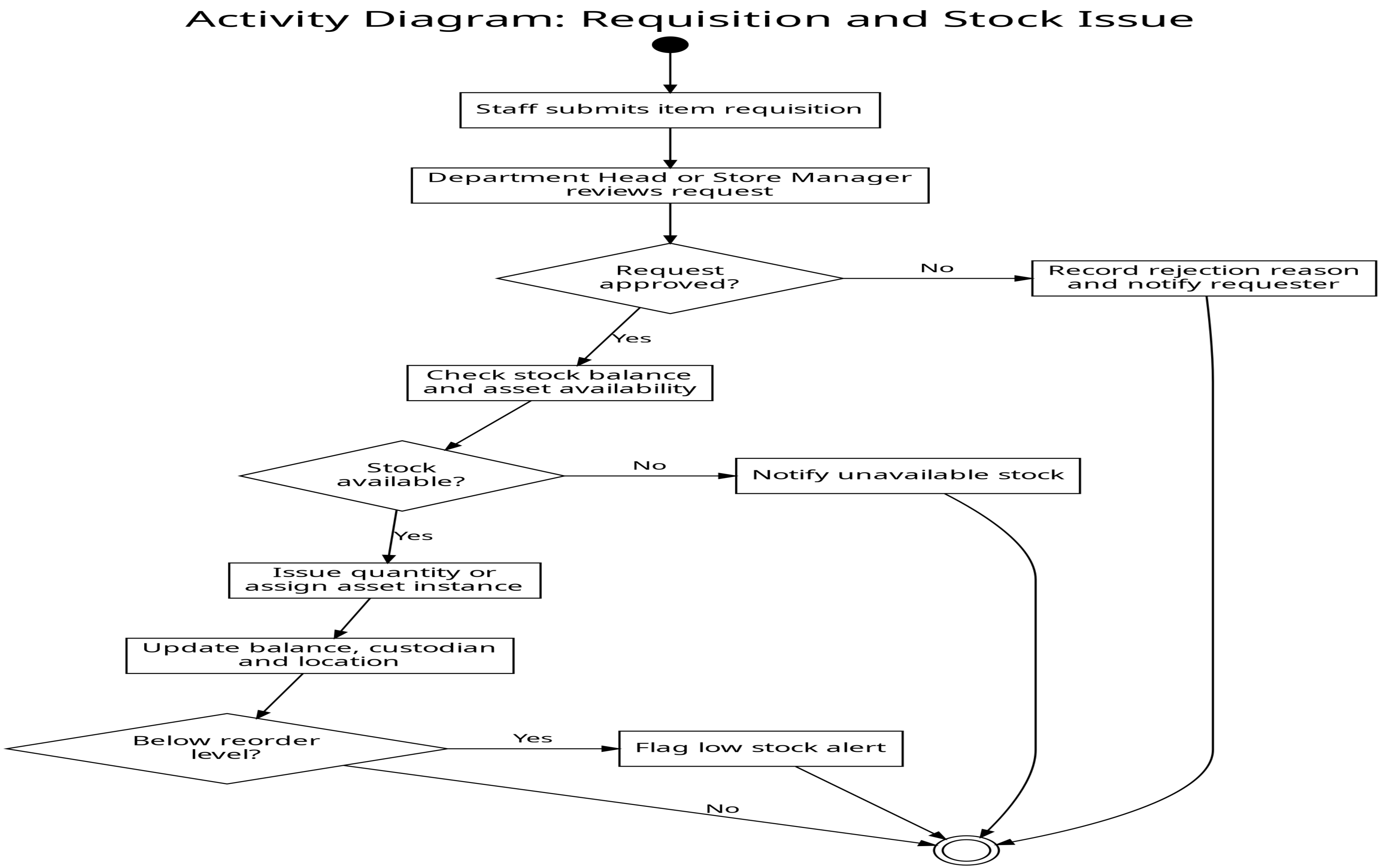


Figure 5 Activity Diagram for Requisition and Stock Issue

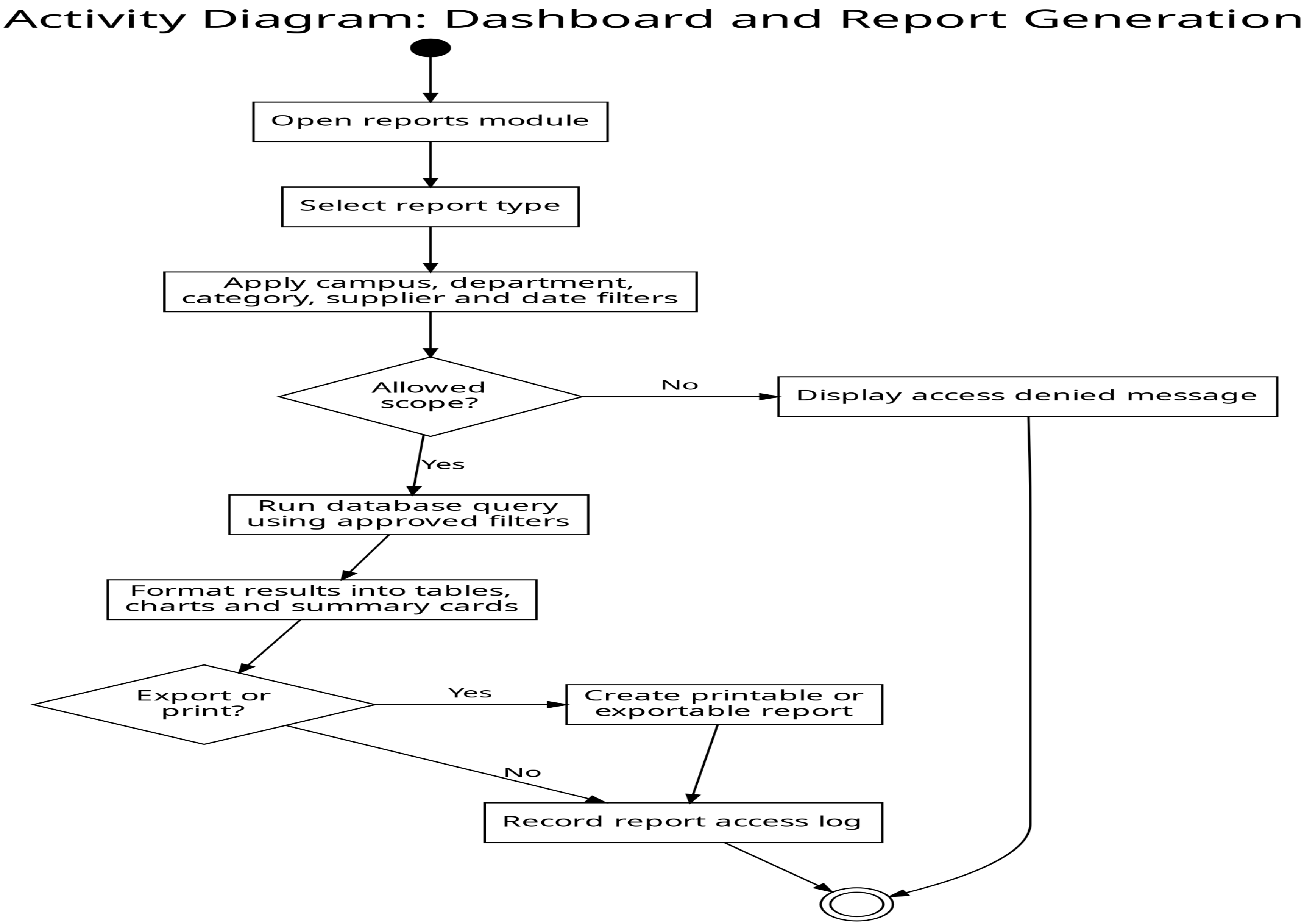


Figure 6 Activity Diagram for Dashboard and Report Generation

7.0 UML Class Design

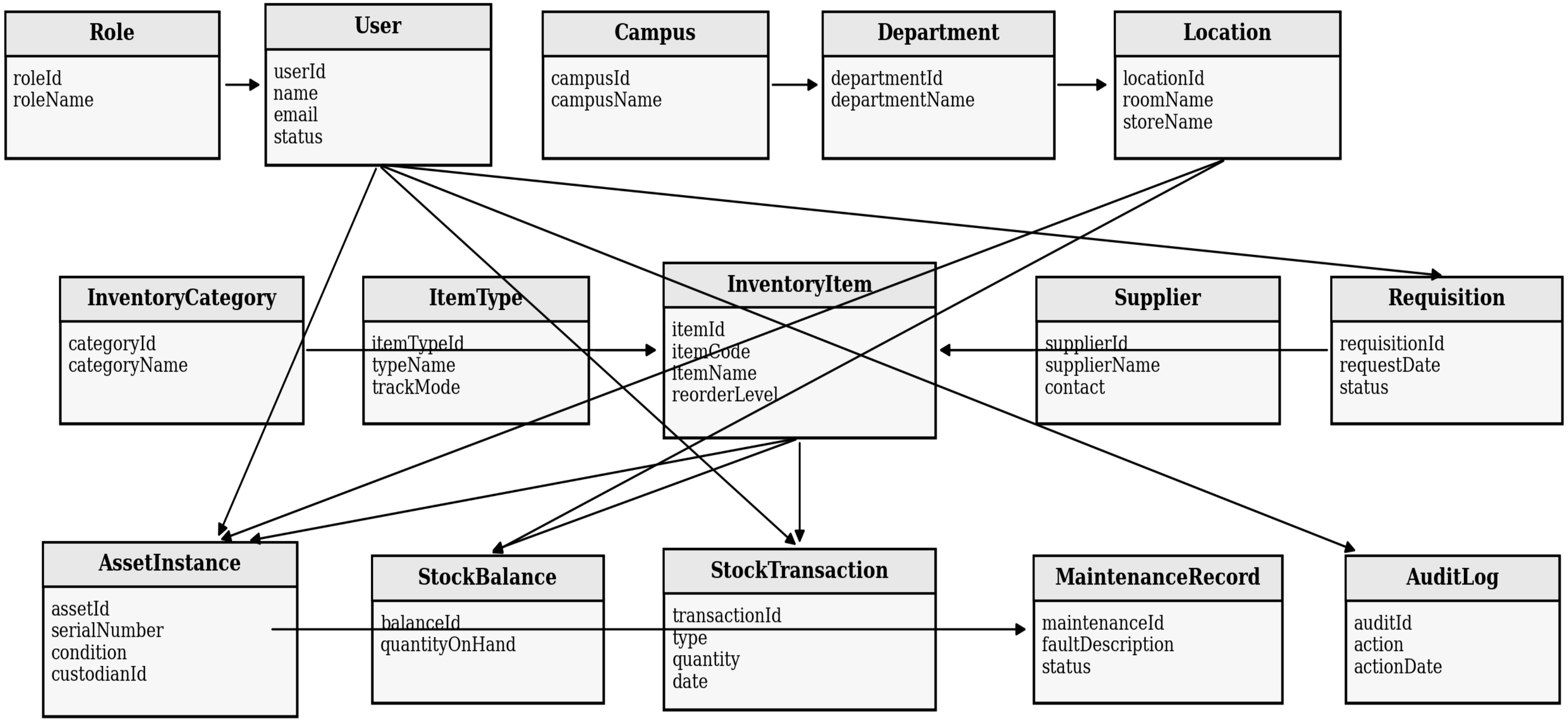
The UML class diagram represents the main software objects that will support the inventory system. It mirrors the database model while adding behaviour such as login, item registration, stock movement, request approval, report access and audit logging.

The class diagram remains at system design level rather than replacing the detailed database design. Attributes and methods are summarized so that the figure can remain legible on one landscape page. The full table definitions are contained in the Database Design Document.

Table 9 Class Model Design Notes

Design Concern	How the Class Model Handles It
Users and Permissions	User is associated with Role so that access can be controlled by responsibility
Physical Organization	Campus, Department and Location represent where inventory is held
Item Catalogue	InventoryItem, ItemType and InventoryCategory describe item identity and tracking mode
Fixed Assets	AssetInstance separates individually traceable assets from general item definitions
Stock Control	StockBalance and StockTransaction maintain available quantities and movement history
Accountability	AuditLog records important user actions for later review

UML Class Diagram for the UIRI Inventory Management System



Arrows show the main class dependency direction. The detailed data fields, keys and constraints are documented in the Database Design Document.

Figure 7 UML Class Diagram for the UIRI Inventory Management System

7.1 Class Responsibility Summary

Table 10 Class Responsibility Summary

Class	Main Responsibility
User and Role	Authenticate users and enforce role based permissions
Campus, Department and Location	Represent the organizational and physical placement of inventory
InventoryItem, ItemType and InventoryCategory	Represent the item catalogue and distinguish fixed assets from consumables
Supplier	Maintain supplier identity and contact details
AssetInstance	Track individual durable assets using serial number, location, status and custodian
StockBalance and StockTransaction	Maintain stock quantities and record all stock movement history
Requisition	Support departmental requests before stock is issued
MaintenanceRecord	Record faults, repairs and maintenance activities for assets
AuditLog	Preserve accountability for user actions

8.0 Design Traceability

Table 11 Design Traceability Matrix

Design Area	Relevant Diagram	Supported Requirement Area
Authentication and access control	Use case diagram, login activity diagram and class diagram	User login, role management, account security and session control
Inventory catalogue and assets	Use case diagram and class diagram	Item registration, category management, fixed asset tracking and condition updates
Stock movement	Stock receipt activity diagram, stock issue activity diagram and class diagram	Stock in, stock out, transfers, returns and stock balance updates
Supplier records	Use case diagram and class diagram	Supplier registration, supplier history and acquisition records
Reports and dashboard	Report generation activity diagram and use case diagram	Dashboard summaries, low stock reports, asset reports and filtered searches
Audit and accountability	Use case diagram and class diagram	Audit logs, report logs and traceability of sensitive user actions

9.0 Conclusion

This document has presented the system design diagrams for the UIRI Inventory Management System. The use case model identifies the actors and system services. The activity diagrams describe the main workflows for login, stock receipt, stock issue and reporting. The class diagram provides a structured object model that supports the database driven implementation.

The revised landscape layout improves figure visibility and makes each diagram easier to review, print and redraw in Microsoft Visio. The next deliverable in sequence is the source code and functional system prototype, unless the supervisor first requires the Visio versions of the diagrams to be refined separately.

10.0 References

[1] ICT Team, UIRI Namanve, "Internship Project Question: Design and Development of an Inventory Management System for Uganda Industrial Research Institute (UIRI)," Uganda Industrial Research Institute, Namanve, Uganda, Project Brief, 2026.

[2] K. P. Kato, K. Tracy and A. D. Kamate, "System Proposal: Design and Development of a Web Based Inventory Management System for UIRI," Uganda Industrial Research Institute, 2026.

[3] K. P. Kato, K. Tracy and A. D. Kamate, "Software Requirements Specification for the UIRI Inventory Management System," Uganda Industrial Research Institute, 2026.

[4] K. P. Kato, K. Tracy and A. D. Kamate, "Database Design Document for the UIRI Inventory Management System," Uganda Industrial Research Institute, 2026.

[5] I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson, 2016.

[6] Object Management Group, "Unified Modeling Language Specification," OMG, 2017.